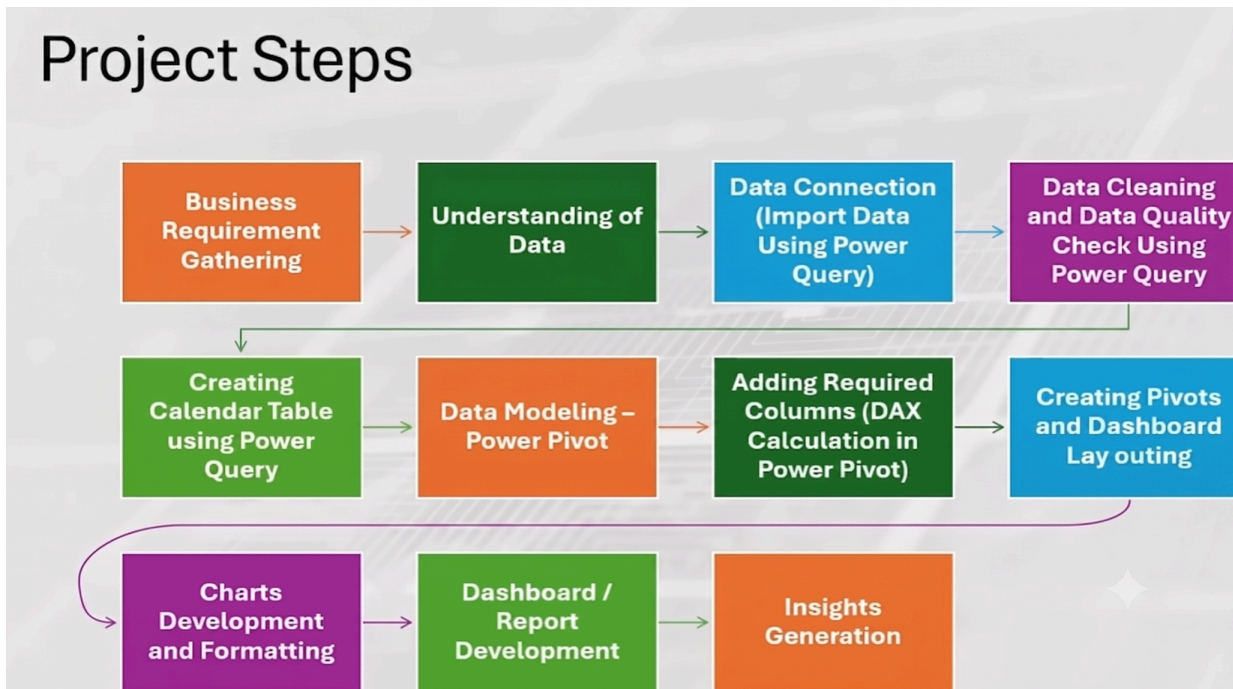


Hospital Emergency Patients Analysis

- End-to-End project using Power Query and Power Pivot

1. Project Requirements



2. Purpose of Project:

- We need to create a ***Hospital Emergency Room Analysis Dashboard*** to improve efficiency and provide useful insights. This dashboard will help Senior stakeholders in monitoring and better decision making for managing patients and improving services.

3. KPIs requirements:

- **1. Number of Patients** : Count total number of patients visiting each day.

- **2. *Average wait time*** : Find the average time that patients waited to see a medical professional.
 - use an area sparkline to track daily changes and highlight days with longer wait times that might need improvements.
- **3. *Patient satisfaction Score***: Check the average daily satisfaction score of patients to assess service quality.
 - use an area sparkline to show trends, spot drops in satisfaction and link them with challenges.

4. Charts to Create

- Patient Admission Status
- Patient Age distribution
- Timeliness of observability (% of patients seen within 30 mins)
- Gender Analysis
- Department Referrals (which department patients are referred to the most)

5. Power Query Editor and Power Pivot (Add-ins)

- Data Transformations and Modelling

1. ***Patient Admission Time*** column splitted into 2 column : **Patient Admitted Date** and **Patient Admitted Time**.

2. **Patient Initial Name** and **Patient Last Name** column merged into one as '**Patient Name**'.

3. **Patient Gender** column has low cardinality values (M, Male, Female, F) → replaced with ('Male', 'Female') for better data quality.

4. **Patient Admission Flag** values changed from ('false' , 'true') → ('Not Admitted' , 'Admitted') respectively.

5. **Patient Admission Flag** column repeated hence removed one occurrence.

6. Added one extra table as '**Date**' using 'DAX' to get continuous dates for better date/time interpretations as '**Patient Admission Date**' column had non-continuous dates.

- List.Dates(#date(2023,01,01),731,#duration(1,0,0,0) → taken 731 total days starting from 1 April,2023 and with a period of '1'.

7. Loaded data as 'Connection' so that larger data should not be crashed while loading into the local system's memory.

8. Entity Relationship created between '**Hospital Emergency Room Data**' table and

'Date' table → Many-to-one relationship
(Hospital → Date)

9. Created **'Age Group'** column using 'IF' formula using DAX

- Age > 70 → 71-80
- Age > 60 → 61-70
- Age > 50 → 51-60
- Age > 40 → 41-50
- Age > 30 → 31-40
- Age > 20 → 21-30
- Age > 10 → 11-20
- Age <= 10 → 00-10

10. Created **'Patient Attend Status'** as:

- Patient Waittime > 30 → Delay
- Patient Waittime <=30 → On-time

6. Learning and Insights:

- Analysis of Patients in the emergency ward on their **Age Group, Gender, Satisfaction score, Admittance/Non-Admittance, Average Wait Time, Referred Department**.
- Above insights, helps Hospital Management to keep actual records of their patients, timely and data-driven decision making.
- It is being observed that **'Average Wait Time'** in emergency wards is around **'30 minutes'**, while in General OPD wards, it usually lies between **40-50 minutes**.

- Generally, most patients were **not referred** to any specific department within a year and cured within emergency settings of the hospital, which says that hospital emergency is well maintained and taken care of.
- * Dashboard setup in **MS-Excel** is much more time consuming than **Power BI** due to its less automatic format style settings. Dashboarding was extremely time consuming and exploring.

THANK YOU :)